	Case Name: **Generic Document Output System**	Sector	**IT**
OR-AS Operations Research Applications and Solutions	**OR-AS** Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Submitted by	**Matilde Casado and Margarida Antunes**		
Date	**2015**	Project Control	Automatic tracking Tracking based on user input
File Name	**C2015-16 Generic Document Output System**		

1. Project description

Project authenticity

The project involves analysis, design, development, testing, and implementation to optimize document processing and system integration.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	30
Planned Duration (PD)	270 days*
Budget At Completion (BAC)	64.620 €
Renewable Resources	13
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	10%
Activity Distribution (AD)	48%
Length of Arcs (LA)	13%
Topological Float (TF)	21%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11.3	17.2	-2.9
CRI-rho	23.9	22.7	-0.9
CRI-tau	36.0	41.7	-0.8

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	6.4	15.3	-3.7
CRI-rho	6.4	15.3	-3.7
CRI-tau	6.4	15.3	-3.7

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	13.3	34.6	-2.3
SI	15.0	25.2	-2.8
SSI	4.2	18.3	-4.9
CRI-r	10.5	18.0	-3.9
CRI-rho	23.4	23.5	-1.2
CRI-tau	37.1	41.7	-0.8

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	107.0	-84.8
PV - SPI	32.2	18.5
PV - SCI	32.1	19.1
ED - 1	353.0	23.5
ED - SPI	32.2	19.3
ED - SCI	32.8	18.5
ES - 1	92.8	-92.1
ES - SPI(t)	32.3	17.8
ES - SCI(t)	32.3	17.8

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	3.1	0.4
CPI	2.9	0.4
SPI	14.6	14.6
SPI(t)	15.8	15.8
SCI	14.6	14.6
SCI(t)	15.8	15.8
0.8 CPI + 0.2 SPI	9.5	9.5
0.8 CPI + 0.2 SPI(t)	12.6	12.6

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 6 tracking periods with no defined length. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	28.8	-9573.7	-399.2	0.9	0.6	0.6	0.7
std dev	1487.7	8319.4	508.3	0.2	0.3	0.5	0.3
final	495.0	0.0	0.0	1.0	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	270 days	Real Duration	270 days	Late	0%

EAC(t)			**Real Accuracy**	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	290.3	1.0	1.2	-1.2
PV - SPI	291.0	1.3	1.2	-1.2
PV - SCI	291.7	1.6	1.3	-1.3
ED - 1	292.5	1.9	1.3	-1.3
ED - SPI	293.3	1.6	1.4	-1.4
ED - SCI	294.0	1.3	1.4	-1.4
ES - 1	294.7	1.0	1.5	-1.5
ES - SPI(t)	295.3	1.0	1.5	-1.5
ES - SCI(t)	296.0	1.3	1.5	-1.5

2.3.3.3. Cost forecasting

BAC	64.620 €	Real Cost	64.125 €	Under Budget	-0.8%

EAC			**Real Accuracy**	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	64591.2	1487.7	0.1	-0.1
CPI	76114.7	23389.0	3.1	-3.1
SPI	153673.5	106332.5	23.3	-23.3
SPI(t)	4416925.9	6776565.3	1131.3	-1131.3
SCI	209638.9	201835.8	37.8	-37.8
SCI(t)	7112336.8	10922274.5	1831.9	-1831.9
0.8 CPI + 0.2 SPI	82656.9	29573.3	4.8	-4.8
0.8 CPI + 0.2 SPI(t)	85068.9	35810.7	5.4	-5.4